

Solving max./min. problems

(1)

The 2nd Derivative Test (RECALL)

① Find the critical numbers of $f'(x)$

(ie: $f'(c) = 0$)

② Substitute the critical numbers into $f''(x)$.

→ If $f''(c) > 0 \Rightarrow$ local minimum at c .

→ If $f''(c) < 0 \Rightarrow$ local maximum at c .

→ If $f''(c) = 0 \Rightarrow$ no conclusion.
use chart on $f'(x)$.

Optimization – Numbers

Example 1

Determine two positive numbers with a product of 450 such that the sum of one of the numbers and twice the second number is as small as possible.

$$\begin{aligned}xy &= 450 \\y &= \frac{450}{x} \\M &= x + 2y \\M &= x + \frac{900}{x} \\M' &= 1 - \frac{900}{x^2}\end{aligned}$$
$$\begin{aligned}1 - \frac{900}{x^2} &= 0 \\x^2 - 900 &= 0 \\x &= 30 \\\Rightarrow y &= 15\end{aligned}$$

$\therefore$ The 2 #'s are 30 & 15.

$$M'' = \frac{1800}{x^3}$$
$$M''(30) = \frac{1800}{(30)^3} > 0$$

$\Rightarrow$ min.

optional

Example 2

The sum of two numbers x and y is 150. Determine the two numbers such that the product

x^2y is a maximum.

positive

$$x + y = 150$$
$$y = 150 - x$$

or $x = 150 - y$

product $\rightarrow P = x^2y$

$$P = x^2(150 - x)$$
$$P = 150x^2 - x^3$$
$$P' = 300x - 3x^2$$
$$-3x(x - 100) = 0$$
$$\Rightarrow x = 0 ; x = 100$$

$\uparrow$ inadmissible

$$\therefore y = 50$$

$\therefore$ The 2 #'s are 100 & 50.

$$P'' = 300 - 6x$$
$$P''(100) = -300 < 0$$

optional

2nd D.T. $\Rightarrow$ max.